# Measuring FOMC Policy Tone with Leakage-Controlled Two-Stage LLMs

Jaeho Choi[1], Chaeshick Chung[1], Seyoung Chung[1], Jaewon Kim[1], Yoonsoo Lee[1], and Daemin Park[2]

[1]Department of Economics, Sogang University
[2]JB Financial Group

February 2026

**Abstract**

Empirical macro-finance increasingly relies on text-based measures of central bank communication to quantify monetary policy stance. However, existing approaches often depend on sentence-level segmentation and dictionary-based preprocessing, while encoder-based pipelines face hard input-length constraints that force the truncation of long documents, such as meeting minutes. We address these limitations by fine-tuning a decoder-only large language model on FOMC communications via domain-adaptive pretraining and supervised fine-tuning. Specifically, we propose a leakage-robust, two-stage measurement framework that infers policy tone directly from full-length post-meeting statements, minutes, and press conferences without sentence-level fragmentation. We establish external validity through financial market reactions: conditional on our tone classifications, policy-rate changes and asset prices move in theoretically expected directions, exhibiting stronger alignment than standard FinBERT-style classifiers. Overall, our full-document, decoder-only approach delivers a highly stable and economically coherent measure of policy tone for long-form central bank communications.

**Keywords:** FOMC communication; monetary policy stance; text-as-data; large language models; leakage control
**JEL Codes:** E52; E58; C55; C63.

# 1 Introduction

Central-bank communication is a first-order state variable in macro–finance. Asset prices respond not only to realized policy-rate changes but also to forward guidance, risk assessments, and subtle shifts in the balance of concerns between inflation and real activity. In the United States, the FOMC produces three primary communication artifacts—post-meeting statements, meeting minutes, and press conferences—that differ in timing, length, and informational structure. A central empirical challenge is therefore measurement: how to map long, stylized policy documents into a stable, interpretable, and economically meaningful indicator of policy stance suitable for downstream analysis.

Much of the existing literature relies on sentence-level segmentation, dictionary-based tone indices, or embedding-based classifiers applied to preprocessed text. While these approaches have generated valuable insights, they face two structural limitations. First, sentence-level or chunk-based pipelines only partially exploit document-level structure, especially in lengthy documents such as meeting minutes where policy signals are embedded in narrative context. Second, encoder-style models impose input-length constraints, forcing truncation or ad hoc recombination of long texts, potentially discarding economically relevant information. These design choices are often pragmatic but introduce a gap between the original policy document and the measured stance variable.

This paper proposes a measurement framework that directly addresses these constraints. We fine-tune a decoder-only large language model on FOMC communications using domain-adaptive pretraining (DAPT) and supervised fine-tuning (SFT), and construct a full-document estimator of FOMC policy tone. Rather than mapping raw text directly to a label, we introduce a two-stage design. Stage 1 generates a structured intermediate summary (MSI) that separates (i) policy actions and forward guidance from (ii) macroeconomic conditions and risks. Stage 2 maps this structured representation to an ordinal tone label (*hawkish*, *neutral*, *dovish*) and a short rationale used as an audit trail. By forcing a decomposition between policy content and macro environment, the procedure reduces sensitivity to rhetorical style and enhances interpretability.

A key econometric concern in such a pipeline is that Stage 2 conditions on a generated regressor. If intermediate summaries are produced in-fold by a Stage 1 model trained on the same observations, evaluation can be optimistically biased. We therefore implement cross-fitting (K=4) when generating training summaries, ensuring that Stage 2 is trained on out-of-fold intermediates and that training inputs more closely resemble deployment conditions. Validation and testing follow a strict temporal split (train $<$2019, test $\geq$2022), eliminating look-ahead bias.

Our empirical results show that the leakage-controlled two-stage SFT design improves out-of-time macro-F1 relative to in-fold ablations, with especially large gains for minutes—the longest and most structurally complex documents in the corpus. More importantly, we evaluate economic coherence rather than classification metrics alone. Conditional on our tone labels, S&P 500 returns and realized changes in the effective federal funds rate move in theoretically expected directions, and the separation between *hawkish* and *dovish* events is sharper than under FinBERT-style baselines. This external validation suggests that the model captures economically relevant directional information rather than superficial sentiment.

Our contribution is threefold. First, we construct an auditable, full-document measure of FOMC policy tone using a decoder-only LLM without truncation or sentence-level preprocessing. Second, we introduce a leakage-robust two-stage SFT design with cross-fit intermediate generation, aligning training and deployment environments. Third, we validate the resulting tone measure against financial-market reactions, positioning it as a practical measurement device for macroeconomic and asset-pricing applications.

The remainder of the paper proceeds as follows. Section 3 describes the corpus, labeling strategy, and temporal split. Section 4 details the measurement framework and training design. Section 5 reports model performance and ablation study results. Section 6 evaluates market-response coherence. Subsequent sections use the constructed tone series in standard empirical specifications.

## 2 Related Work

A large literature studies how central bank communication moves asset prices and shapes expectations. Within this literature, the Federal Open Market Committee (FOMC) provides a uniquely rich set of communication channels—post-meeting statements, meeting minutes, and post-meeting press conferences—that enable researchers to connect the content and tone of policy communication to high-frequency market responses.

Early work extracts directional signals from FOMC statements using text-based measures. For example, Lucca and Trebbi (2009) quantify the semantic orientation of statements by operating at the sentence level, constructing search units from part-of-speech-based chunks and aggregating sentence-level measures to a statement-level index. A related strand focuses on topic-content separation: Hansen and McMahon (2016) apply topic modeling to statements using paragraph-level representations (after standard text cleaning such as stop-word removal and stemming) and then measure tone conditional on topic-relevant sentences.

Dictionary-based approaches have also been widely used to build signed measures of policy communication. Gardner et al. (2022) construct an FOMC sentiment index from topic keywords and modifier keywords (and select phrases for forward-looking policy actions), explicitly scaling the index after deleting uninformative sentences. Such Fed-specific dictionaries have subsequently been used to build comparable sentiment measures across multiple textual sources, including statements and press conference transcripts.

Beyond released statements, alternative or counterfactual policy language has been used to distinguish expected from surprise components in communication. Doh et al. (2025) leverage alternative FOMC statements and embedding-based representations of text (with explicit tokenization and vectorization of unstructured text strings) to decompose communication into expected tone and novelty components using high-frequency market data.

Meeting minutes provide longer and more detailed descriptions of policy deliberations than post-meeting statements. Topic-model approaches commonly rely on paragraph-level units and remove boilerplate or operational sections that are not central to deliberation. For instance, Jegadeesh and Wu (2017) construct paragraph-level representations of minutes after excluding specific sections (e.g., introductory and operations-related parts) and applying standard vocabulary cleaning (e.g., stop-word removal) before estimating topic structure. Complementary work studies the information content and market impact of FOMC minutes releases. Rosa (2013) documents significant financial-market responses around minutes announcements, and Jung (2016) shows that minutes can improve markets' ability to predict subsequent target-rate changes. More recently, Tadle (2022) constructs a document-level sentiment measure for minutes and finds that it contains incremental information about future policy moves. Recent evidence also highlights both the potential and the limitations of language models when applied to FOMC minutes (Kim et al., 2023).

Press conferences add a live, partially unscripted channel to FOMC communication. A common choice is to focus on the Chair's answers during the Q&A portion and omit reporters' questions when constructing text measures, reflecting the goal of isolating the policymaker's unscripted communication (De Pooter, 2021). More recent work incorporates non-verbal information from audio/video recordings. Gorodnichenko et al. (2023) split press-conference audio into question–answer segments, focus on the Chair's answers during Q&A, and extract acoustic features (e.g., MFCCs and spectrogram-based representations) to quantify vocal emotions and their market impact.

Recent studies align the timing of verbal communication with high-frequency prices. Gómez-Cram and Grotteria (2022) scrape press-conference videos, convert audio to text, timestamp spoken content, and align the resulting sequence with high-frequency asset prices to study real-time price discovery. Related work emphasizes that language in the Q&A can depart from the post-meeting statement and that this departure is informative for market dynamics. For example, Naraina and Sangani (2026) build word-level valence using market reactions to statements and use it to measure the valence of opening remarks and Q&A language, as well as cosine-similarity departures between Q&A and the statement/minutes.

Because market participants consume policy communication through intermediaries, a growing literature measures sentiment in financial news and social media around announcement days. Schmanski et al. (2023) collect news (via Factiva/Dow Jones) and Twitter posts using keyword-based retrieval, remove duplicate sentences in newswire text, and construct sentiment measures using a Fed-specific dictionary at the sentence level. Their results highlight that media and social sentiment can diverge from the sentiment extracted directly from the FOMC, with implications for yield reactions and expectation formation.

Recent work uses large language models (LLMs) to classify central bank communication at fine granularity. More recently, Hansen and Kazinnik (2024) demonstrate that GPT models, particularly GPT-4, can significantly outperform traditional NLP methods in classifying the policy stance encoded in FOMC communications and provide human-level explanations for their classifications, further showing that large language models can be leveraged to identify macroeconomic shocks using narrative identification strategies without imposing strong parametric assumptions.

Silva et al. (2025) classify individual sentences across multiple dimensions (topic, stance, audience, sentiment) in a multilingual corpus, emphasizing the importance of language-aware sentence segmentation to reduce boundary errors across languages. Kwon et al. (2025) employ a long-context large language model (LLM) to analyze 25 years of financial news and construct sentiment indices for U.S. growth and inflation. Without imposing additional statistical identifying assumptions, the authors further decompose the underlying structural drivers on the demand and supply sides in a narrative framework, and demonstrate that this approach improves high-frequency market monitoring and predictive performance.

We jointly analyze the tone of multiple FOMC communication channels—statements, press conferences, and minutes—using long-context language models that allow us to process full documents without truncation. This full-document approach complements prior work that typically relies on sentence- or paragraph-level representations, excludes selected sections (e.g., focusing only on Q&A answers), or deletes uninformative/duplicate sentences to construct indices. By preserving the entire original text (while maintaining a structured parse for timing and document hierarchy), we enable a unified and potentially real-time mapping from communication tone to contemporaneous financial-market responses.

**FOMC Statements: Text-Based Measures of Stance and Tone**

## 3 Data

We construct a meeting-level dataset spanning 1994–2026 with three primary document types: policy statements, minutes, and press-conference opening statements. For each meeting date, we retain at most one document per type when available. Our corpus therefore includes all documents that are publicly available and retrievable for each meeting within 1994–2026. Press-conference opening statements are available only for meetings with press conferences; hence coverage is lower than for statements and minutes.

Table 1: Corpus coverage (meeting-level).

| **Item** | **# of Meetings** | **Coverage** |
|---|---|---|
| Meetings (1994–2026) | 245 | 100% |
| Policy statements | 245 | 100% |
| Minutes | 225 | 92% |
| Press conferences (openings) | 90 | 37% |

We use two complementary label sources: (i) a statement-canonical, meeting-level tone label series used for broad supervision; and (ii) a smaller press-conference label set (tone + short rationale), used for refinement and press-focused evaluation.

All labels are obtained via LLM-based annotation, but the labeling protocol differs across communication channels.

**Statement and Minutes.** For post-meeting statements and meeting minutes, tone labels are generated using a single high-capacity model, *Claude 4.6 Opus (Thinking)*. These documents contain relatively explicit and structured information about the policy decision. In particular, statements formally announce the target range decision for the federal funds rate, and minutes document the deliberation and justification surrounding that decision. Because the direction of policy (hike, hold, or cut) is comparatively well-defined in these texts, a single strong reasoning model provides stable and internally consistent annotations.

**Press Conferences.** In contrast, press-conference transcripts—especially the opening statement—contain forward-looking guidance, risk assessments, and unscripted clarifications that often convey more nuanced and context-dependent tone signals. To reduce model-specific bias and increase robustness in this more interpretative setting, press-conference tone labels used in the refinement stage are generated independently by four LLMs: *Grok 4.1(Fast Thinking), GPT-5(Thinking), DeepSeek-v3.2(Thinking), and gemini-3-flash-preview(Thinking)*. We aggregate the four outputs via majority voting (Amgalan et al. (2025)); in the event of a tie, we use the label predicted by *Grok 4.1 (Fast Thinking)* as the tie-breaker.

## Data Split

We impose a strictly time-ordered evaluation scheme in which observations dated before 2019-01-01 are assigned to the training set, observations from 2019-01-01 to 2021-12-31 are assigned to the validation set, and observations dated 2022-01-01 or later are assigned to the test set.

All model selection is conducted using only the Train and Validation sets; the Test set is reserved as a final holdout.

Importantly, the temporal split is defined by the FOMC meeting date rather than the document release date. While post-meeting statements and press conference transcripts are released on or shortly after the meeting day, meeting minutes are published with a delay of approximately three weeks. As a result, documents that reflect the same underlying information set may have different public release dates.

To ensure that documents corresponding to the same policy meeting share the same information set and are assigned to the same split, we group all documents by meeting date before partitioning the data. This design prevents artificial performance inflation in the Test set and eliminates potential look-ahead bias during training.

# 4 Measurement Framework (Methodology)

This paper develops a structured measurement framework for central bank communication using large language models (LLMs). We analyze FOMC statements, minutes, and press conferences and construct an economically interpretable measure of policy tone. Unlike prior approaches that rely on dictionary-based sentiment or direct text-to-label mapping, we introduce a two-stage supervised fine-tuning (SFT) design that separates structured policy summarization from tone classification.

Our framework combines domain-adaptive pretraining (DAPT) on FOMC text with supervised fine-tuning and explicit leakage control via cross-fitting. This design substantially improves classification stability, economic coherence, and out-of-sample market alignment. The result is a measurement system that is both auditable and empirically useful for downstream financial and macroeconomic analysis.

## Policy Tone Definition

Let $x_{i,d}$ denote the text associated with meeting $i$ and document type $d \in \{\texttt{statement}, \texttt{minutes}, \texttt{presconf}\}$. Our target variable is an ordinal measure of FOMC policy tone,

$$y_{i,d} \in \{\textit{dovish}, \textit{neutral}, \textit{hawkish}\},$$

which we interpret as the net directional signal about the likely path of the policy rate over the next several meetings (Appendix A). We treat $y_{i,d}$ as a text-derived measurement output designed for downstream empirical analysis.

## Two-Stage Estimation with Structured Summaries

We implement a two-stage estimator that constructs an intermediate structured representation, which we call **Multi-Scale Insight (MSI)** and denote by $z_{i,d}$. The object $z_{i,d}$ is JSON-formatted and contains three fields: `general` (overall policy summary), `subtask` (concrete policy actions and forward guidance), and `environment` (macroeconomic conditions and risks). This decomposition encourages the model to separate policy actions from the broader macroeconomic narrative, reducing sensitivity to rhetorical style while improving auditability. The second stage maps $z_{i,d}$ to a tone prediction $\hat{y}_{i,d}$ and a supporting rationale $\hat{r}_{i,d}$.

## SFT Training Objectives

The model is initialized from a base language model adapted to the FOMC domain (DAPT) and subsequently fits two conditional generators. In **Stage 1 (Summary Training)**, the model $q_\phi(z \mid x)$ is trained to generate the MSI from raw FOMC minutes $x_{i,d}$ via completion-only maximum likelihood. In **Stage 2 (Tone Training)**, the generator $p_\psi(y, r \mid z)$ is trained to predict the synthesized tone and rationale using the generated MSI as input.

**Stage 0: domain-adaptive pretraining (DAPT).** Let $\mathcal{C}$ be the unlabeled FOMC corpus and $u = (u_1, \ldots, u_T)$ a token sequence. We minimize negative log-likelihood:

$$\mathcal{L}_{\text{DAPT}}(\theta) = -\mathbb{E}_{u \sim \mathcal{C}} \left[ \sum_{t=1}^{T} \log p_\theta(u_t \mid u_{<t}) \right]. \tag{1}$$

In practice, updates are implemented with parameter-efficient adapters (e.g., QLoRA/LoRA) on a frozen base model.

**Stage 1: Supervised Summary Training** In this stage, we train the model to generate Multi-Scale Insight(MSI) from the original Federal Open Market Committee (FOMC) minutes. Given training pairs $(x_{i,d}, z^\star_{i,d})$, where $x_{i,d}$ represents the raw text of the FOMC document and $z^\star_{i,d}$ is the target summary, we fit the summary generator by completion-only maximum likelihood:

$$\mathcal{L}_{\mathrm{MSI}}(\phi) = -\mathbb{E}_{(x,z^\star)\sim\mathcal{D}_{\mathrm{MSI}}}\left[\sum_{t=1}^{|z^\star|}\log q_\phi(z^\star_t \mid x, z^\star_{<t})\right]. \tag{2}$$

**Stage 2: Supervised Tone Training** Given the generated summaries (MSI) $z_{i,d}$ from Stage 1, we train the model to map these summaries to their corresponding target outputs. Specifically, given training pairs $(z_{i,d}, a^\star_{i,d})$, where $a^\star_{i,d}$ is a strict JSON completion containing the synthesized tone $y^\star$ and a brief supporting rationale, we minimize the following objective:

$$\mathcal{L}_{\mathrm{tone}}(\psi) = -\mathbb{E}_{(z,a^\star)\sim\mathcal{D}_{\mathrm{tone}}}\left[\sum_{t=1}^{|a^\star|}\log p_\psi(a^\star_t \mid z, a^\star_{<t})\right]. \tag{3}$$

We train in two phases: a broad warmup phase (all document types with statement-canonical labels) followed by refinement on press-conference labels.

### Leakage Control and Robustness

To address the identification issue where Stage 2 conditions on a generated regressor $z_{i,d}$, we implement a **cross-fitting** strategy to mitigate data leakage. We partition the training meetings $\mathcal{I}_{\mathrm{tr}}$ into $K = 4$ folds, denoted as $\{\mathcal{I}^{(k)}\}_{k=1}^K$; for each fold $k$, the Stage 1 model is trained on $\mathcal{I}_{\mathrm{tr}} \setminus \mathcal{I}^{(k)}$ to generate summaries for $\mathcal{I}^{(k)}$, ensuring that downstream estimators do not face an artificially optimistic input distribution. To further regularize Stage 2 against minor phrasing shifts, we generate multiple summary variants for each training document using a fixed decoding grid (one greedy plus four sampled variants), while maintaining deterministic greedy decoding for validation and test sets to ensure stability. We evaluate our approach through three primary estimators: (i) TWO-STAGE (CROSS-FIT), our preferred design with cross-fit summaries; (ii) TWO-STAGE (IN-FOLD), an ablation prone to leakage using in-fold summaries; and (iii) TEXT-ONLY, which maps documents directly to tone as a text-only baseline.

## 5 Empirical Results

We report accuracy and macro-F1 (primary) due to class imbalance and regime shift. All reported test performance is on the strict 2022+ holdout.

TWO-STAGE (CROSS-FIT) achieves the highest overall macro-F1 (0.7270) and a modest accuracy gain relative to TWO-STAGE (IN-FOLD) and TEXT-ONLY. We emphasize macro-F1 as the primary metric because the test set is class-imbalanced (HAWKISH $N$=48, NEUTRAL $N$=33, DOVISH $N$=18) and because the intended use is to construct a stable policy-tone measure rather than to maximize the majority-class hit rate.

The gains from cross-fitting are concentrated in *minutes*, which are the longest and most heterogeneous source. This pattern is consistent with a measurement interpretation: in-fold

Table 2: Out-of-time test performance (2022+, $N = 99$). Entries report macro-F1 with accuracy in parentheses.

| | TWO-STAGE (IN-FOLD) | **two-stage (cross-fit)** | TEXT-ONLY |
|---|---|---|---|
| Overall ($N = 99$) | 0.6854 (0.7273) | **0.7270** (0.7475) | 0.6817 (0.7172) |
| Statement ($N = 34$) | 0.7479 (0.7647) | **0.7644** (0.7647) | 0.7494 (0.7353) |
| Press conf. ($N = 33$) | **0.7093** (0.7576) | 0.6988 (0.7576) | 0.6830 (0.7576) |
| Minutes ($N = 32$) | 0.5833 (0.6562) | **0.7013** (0.7188) | 0.5929 (0.6562) |

generation produces a more "in-sample" intermediate distribution that does not generalize as well to later periods and longer documents.

A per-class table is reported in Appendix Table A.1. The in-fold variant exhibits a pronounced *neutral collapse* (high precision but low recall for neutral), which can mechanically create spurious tone swings by mapping ambiguous meetings to extreme categories. Cross-fitting improves neutral recall while maintaining strong hawkish performance. The text-only baseline achieves the highest neutral recall but under-detects dovish episodes, which can distort the tone measure during easing phases.

# 6 Financial Market-Response Validation Design

### Federal Funds Rate Alignment

We examine whether the tone classifications produced by different language models are aligned with realized monetary-policy outcomes around FOMC meetings. For each scheduled meeting date $t$, we merge the event calendar with the effective federal funds rate (EFFR) from FRED (`FEDFUNDS`) and construct a meeting-to-meeting change measure,

$$\Delta\text{EFFR}_t = \text{EFFR}_t - \text{EFFR}_{t^-},$$

where $t^-$ denotes the previous meeting. This statistic captures realized tightening or easing between consecutive FOMC decisions.

Using the held-out test set, we compare three tone estimators: (i) our FOMC-tailored decoder-only model, (ii) the generic `ProsusAI/FinBERT`, and (iii) `ZiweiChen/FinBERT-FOMC`. For each model, we map its output into a three-way tone label (Hawkish, Neutral, Dovish) and compute the distribution of $\Delta\text{EFFR}_t$ conditional on these labels.

A time-series plot of the EFFR level with tone markers from our model reveals a clear qualitative pattern: meetings labeled Hawkish cluster in tightening phases of the cycle, Neutral labels dominate plateau periods, and Dovish labels appear during easing episodes. This visual evidence suggests a systematic mapping between our tone classifications and realized policy direction.

We then formalize this comparison by computing the mean meeting-to-meeting change in EFFR conditional on each tone category and model. The results show a striking monotonic ordering for our estimator. The mean $\Delta$EFFR is strongly positive for meetings classified as Hawkish, close to zero for Neutral, and clearly negative for Dovish. The Hawk–Dove gap exceeds 50 basis points on average, indicating economically meaningful separation across tone buckets. Formally,

$$\mathbb{E}[\Delta\text{EFFR} \mid \text{Hawkish}] > \mathbb{E}[\Delta\text{EFFR} \mid \text{Neutral}] > \mathbb{E}[\Delta\text{EFFR} \mid \text{Dovish}],$$

with substantial magnitude differences.

By contrast, the two FinBERT baselines fail to exhibit such directional consistency. In particular, FinBERT assigns a positive average ΔEFFR even to meetings labeled Dovish, implying that its notion of dovishness is not tightly linked to realized easing steps. Similarly, FinBERT-FOMC does not produce a stable monotonic mapping between tone categories and subsequent policy-rate changes. In several cases, meetings classified as Dovish by the baselines are followed by further rate increases; see Figure 1.

Taken together, these results indicate that our FOMC-specific decoder-only model captures policy-relevant directional content rather than generic financial sentiment. Conditional on our tone labels, realized changes in the policy rate move in theoretically expected directions with sharper separation than under baseline classifiers. This alignment supports interpreting the constructed tone series as an economically coherent proxy for systematic monetary-policy stance.

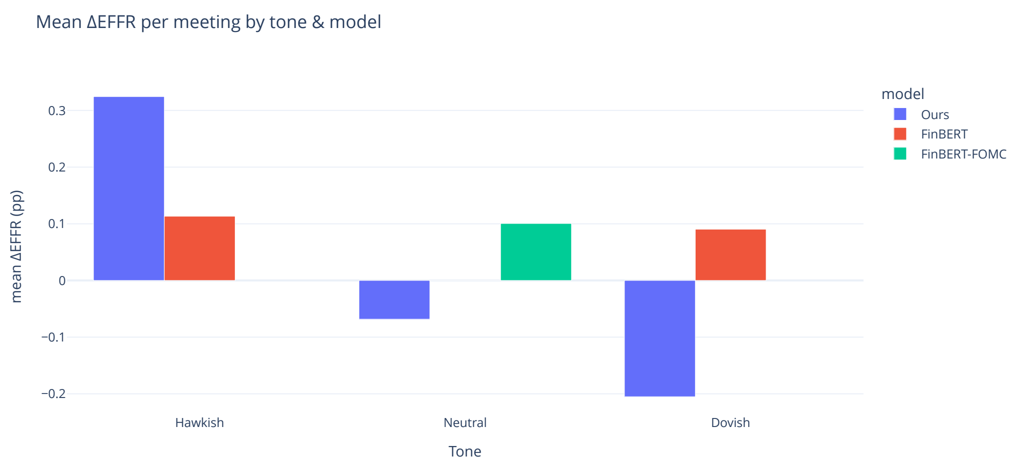


Figure 1: Meeting-to-Meeting Change in the Effective Federal Funds Rate (ΔEFFR) by Tone Classification.

*Notes:* The figure plots, for each tone category, the average change in the effective federal funds rate (EFFR) at FOMC meetings across three tone models (Ours, FinBERT, FinBERT-FOMC). The x-axis shows tone labels (Hawkish, Neutral, Dovish), while the y-axis reports the mean meeting-to-meeting change in EFFR in percentage points. Bars above zero indicate meetings that were, on average, followed by rate hikes, whereas bars below zero indicate rate cuts. Comparing models within each tone category illustrates how closely each model's tone classification aligns with the realized direction of monetary policy.

## FX Market

The FX results reinforce the interpretation that our tone measure captures policy-path expectations rather than generic sentiment. Under standard uncovered interest parity intuition, hawkish signals should raise expected interest-rate differentials and lead to dollar appreciation, while dovish signals should imply depreciation. Our classifier exhibits precisely this pattern: Hawkish meetings are followed by positive mean DXY changes, and Dovish meetings by negative changes. In contrast, FinBERT produces a positive average dollar response even under Dovish classifications, suggesting that its tone categories are less tightly linked to revisions in expected policy paths. This divergence indicates that our FOMC-specific training design more effectively captures the signaling channel of monetary policy.


Figure 2: Five-Day Change in the U.S. Dollar Index (DXY) by Policy Tone and Model. The figure reports the mean 5-day change in DXY around FOMC meetings conditional on tone classifications (Hawkish, Neutral, Dovish) from three models (Ours, FinBERT, FinBERT-FOMC). Positive values indicate dollar appreciation, while negative values indicate depreciation. Our classifier shows stronger alignment with the expected relationship between policy tone and dollar movements, particularly for Dovish events.

# 7 Conclusion

This paper develops an auditable measurement framework for central bank communication and constructs a document-level series of FOMC policy tone. In contrast to sentence-segmented approaches or encoder pipelines that rely on truncation, we fine-tune a decoder-only large language model using domain-adaptive pretraining (DAPT) and a structured two-stage supervised design. By processing full documents—including long meeting minutes—and introducing cross-fitted intermediate summaries, the proposed framework reduces information loss and mitigates optimistic evaluation arising from generated regressors. Empirically, the resulting tone measure exhibits strong external coherence. Conditional on our *hawkish/neutral/dovish* classifications, equity returns and realized policy-rate changes move in theoretically expected directions, with sharper separation than FinBERT-style baselines. This evidence indicates that the model captures economically meaningful directional content rather than surface linguistic sentiment.

Across market-based validation exercises, the constructed tone series remains economically coherent and interpretable, supporting its use as a policy-stance measurement device rather than a generic sentiment signal.

Taken together, the findings suggest that a full-document, decoder-only SFT framework can provide a stable and economically coherent measurement device for central bank communication. More broadly, the paper illustrates how careful training design, leakage control, and market-based validation can transform large language models from generic text classifiers into disciplined tools for empirical macro–finance research.

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# Appendix A Label Definitions

We operationalize tone as *net directional bias about the likely policy-rate path over the next few meetings*. Below is the prompt used for labeling.

- **HAWKISH:** bias toward higher rates or maintaining restrictive rates longer ("high for longer").
- **DOVISH:** bias toward easing/cuts becoming more likely.
- **NEUTRAL:** balanced/no clear directional bias in the policy path.

A "pause" can be *hawkish* if guidance implies holding restrictive policy longer than expected or hiking again.

**Prompt template used for MSI generation and policy-tone labeling (Python)**

```
SYSTEM_PROMPT = """You are an expert central bank watcher and monetary policy analyst.

Your task has TWO stages:
1. FIRST: Generate structured insights (MSI) from the FOMC statement
2. THEN: Based on the MSI, classify the policy stance

=== STAGE 1: MSI Generation ===

Extract three types of insights from the statement:

**General Insight**: A single concise sentence capturing the central theme of the monetary
    policy stance(Goal, Economic approach).

**Subtask Insights** (2-3 sentences):
- Specific actions or decisions with exact numbers/percentages if mentioned
- Forward guidance or conditional commitments regarding future policy actions.
- E.g., interest rate changes, asset purchase decisions, specific policy actions

**Environment Insights** (3-4 sentences):
- Current macroeconomic conditions: inflation rates, unemployment, GDP growth
- Economic or financial risks/challenges
- Key economic indicators mentioned

=== STAGE 2: Labeling based on MSI ===

### Definition (Policy-path stance):
The net directional bias communicated about the likely future path of the policy rate over the
    next several meetings.

(Note: A decision to "Pause" (hold rates) can still be Hawkish if the bias is toward holding
    for longer than expected or hiking again.)

### CRITICAL INSTRUCTIONS:

1. **Ignore Backward-Looking Descriptions:** Do NOT classify based on descriptions of current
    high inflation or strong labor markets alone. High inflation does not automatically mean
    Hawkish if the Fed signals they are done hiking.

2. **Apply Weighting:**
   - **High Weight:** Explicit forward guidance ("expects", "will", "prepared to", "appropriate
     to maintain").
```

```
   - **Medium Weight:** Risk balance ("risks to inflation are upside", "attentive to shortfall").
   - **Low Weight:** Generic boilerplate ("data dependent", "proceeding carefully").

3. **Resolve Conflicts:** If the economic assessment sounds bad (e.g., high inflation) but the guidance is passive (e.g., "we can afford to wait"), prioritize the GUIDANCE.

### Classification Rubric:

**1) Hawkish (Tightening Bias / High-for-Longer)**
- Net bias toward higher rates OR maintaining restrictive rates for a longer duration.
- Key evidence: Pushing back against cut expectations; signaling hikes are still possible; emphasizing that inflation is the dominant risk that prevents easing.

**2) Dovish (Loosening Bias)**
- Net bias toward easing / cuts becoming more likely.
- Key evidence: Openness to cuts in the near term; emphasis on downside growth/employment risks; signaling that the current stance might be too restrictive.

**3) Neutral (Balanced / No Bias)**
- The stance is truly balanced with no clear tilt.
- Key evidence: Emphasizes that the current level is "well-positioned" to handle risks in EITHER direction; explicitly states risks are balanced; avoids signaling the next move.

=== OUTPUT FORMAT ===

IMPORTANT: Generate the ’reasoning’ field BEFORE the ’label’ to ensure logical consistency.
Output ONLY valid JSON with no additional text:

{
  "msi": {
    "general": "A concise summary of the main policy stance",
    "subtask": ["Specific action 1 with numbers", "Specific action 2"],
    "environment": ["Economic condition 1", "Risk/challenge 1", "Key indicator 1"]
  },
  "keywords": ["keyword1", "keyword2", "keyword3", "keyword4"],
  "reasoning": "Brief explanation based on MSI why this label was chosen",
  "label": "Hawkish",
  "confidence": 85
}

Note:
- "label" must be exactly one of: "0", "1", "2"
- Each labeled number means the following:
  - 0: Hawkish
  - 1: Dovish
  - 2: Neutral
- "confidence" must be an integer from 0 to 100
- "keywords" must contain 4 key terms that support the label decision
"""
USER_PROMPT_TEMPLATE = """Analyze the following FOMC Chair’s Opening Statement and provide MSI (structured insights) and policy stance classification.

=== FOMC STATEMENT ===
{statement}
=== END OF STATEMENT ===
```

```
Generate your analysis in the specified JSON format."""
```

# Appendix B Prompts and Schemas

We use the following Stage 2 prompt template for the student model.

**Stage 2 prompt template (student model)**

```
STAGE2_PROMPT_TEMPLATE = """Task: classify the policy-path tone from the structured summary
    below.
Return JSON only:
{"tone":"DOVISH | NEUTRAL | HAWKISH","rationale":"1-2 sentences"}

Definitions:
- HAWKISH = bias to higher rates or longer restrictive policy
- DOVISH = bias toward easing / cuts
- NEUTRAL = balanced / no clear directional bias

DOC_TYPE: {statement|presconf|minutes}
MSI: {...}
"""
```

# Appendix C Additional performance breakdowns

In this section we report detailed model performance.

Table A.1: Per-class performance on the 2022+ test set ($N = 99$). Each cell reports **F1** with (Precision/Recall) in parentheses.

| Model | DOVISH ($N = 18$) | NEUTRAL ($N = 33$) | HAWKISH ($N = 48$) |
|---|---|---|---|
| TWO-STAGE (IN-FOLD) | **0.8718** (0.8095 / **0.9444**) | 0.3810 (**0.8889** / 0.2424) | 0.8034 (0.6812 / **0.9792**) |
| **two-stage (cross-fit)** | 0.7647 (0.8125 / 0.7222) | 0.5862 (0.6800 / 0.5152) | **0.8302** (0.7586 / 0.9167) |
| TEXT-ONLY | 0.5926 (**0.8889** / 0.4444) | **0.6316** (0.5581 / **0.7273**) | 0.8211 (**0.8298** / 0.8125) |